

PAPER- 2 : MB-402: ENVIOREMENTAL MICROBIOLOGY

1.WRITE SHORT NOTE ANY FIVE QUESTION.

(i) Food Web.?

Ans-A food web is a network of interconnected food chains in an ecosystem, showing how different organisms are related through their feeding relationships. It illustrates **who eats whom** and how energy and nutrients flow from one organism to another.

Key points:

- **Producers:** Usually plants or algae that make their own food through photosynthesis.
- **Consumers:** Organisms that eat others; include herbivores (plant-eaters), carnivores (meat-eaters), and omnivores (both).
- **Decomposers:** Organisms like fungi and bacteria that break down dead matter, recycling nutrients back into the ecosystem.
- **Importance:** Maintains ecological balance, shows energy transfer, and highlights the interdependence of species.

(ii) Energy transfer efficiency.?

Ans- Energy Transfer Efficiency measures how effectively energy is converted from one form to another. It is calculated as:

$$\text{Efficiency (\%)} = \frac{\text{Useful Energy Output}}{\text{Total Energy Input}} \times 100$$

High efficiency means less energy is wasted (e.g., heat, sound). Examples: electric motors (~90%), LEDs (~80–90%).

(iii) Vertical Profiles.?

Ans- Vertical Profiles-A **vertical profile** refers to the variation of a physical, chemical, or biological parameter with **height or depth** in the atmosphere, ocean, or soil. It helps understand how environmental conditions change vertically.

Key Points:**1. Atmosphere:**

- Measures temperature, humidity, pressure, wind speed, and composition at different altitudes.
- Example: **Temperature lapse rate** — temperature usually decreases with height in the troposphere.

2. Oceanography:

- Shows how temperature, salinity, and density change with depth.
- Example: **Thermocline** — a layer where water temperature drops rapidly with depth.

3. Soil:

- Vertical profiles describe moisture, nutrients, or organic matter at different soil depths.

Importance:

- Helps in **weather prediction, climate studies, ocean circulation analysis**, and **agriculture**.
- Reveals **layers and stability** in atmosphere and oceans (e.g., inversion layers, stratification).

(iv) Algae toxins.?**Ans- Algae Toxins (Harmful Algal Blooms – HABs):**

- **Definition:** Toxic compounds produced by certain algae, especially during rapid algal growth known as *harmful algal blooms*.
- **Common Types of Algae Producing Toxins:**
 - **Cyanobacteria (blue-green algae)** – freshwater blooms.
 - **Dinoflagellates** – marine environments (cause red tides).
- **Major Toxins:**

1. **Microcystins** – liver toxins from cyanobacteria.

2. **Saxitoxins** – neurotoxins causing paralytic shellfish poisoning.
3. **Domoic acid** – neurotoxin causing amnesic shellfish poisoning.
4. **Okadaic acid** – causes diarrhetic shellfish poisoning.
 - **Effects on Humans and Animals:**
 - Skin irritation, gastrointestinal illness, liver damage, neurological symptoms, or even death in severe cases.
 - **Environmental Impact:**
 - Fish kills, disruption of aquatic ecosystems, contamination of drinking water.
 - **Prevention/Control:**
 - Reduce nutrient runoff (nitrogen, phosphorus), monitor water bodies, avoid drinking or swimming in contaminated water.

(v) Biofouling.?

Ans- Biofouling is the undesirable accumulation of microorganisms, plants, algae, and animals on submerged surfaces, especially on ships' hulls, pipelines, and marine structures. It begins with the formation of a **biofilm** of bacteria, followed by attachment of larger organisms like barnacles, mussels, and seaweeds.

Effects:

- Increases **drag** on ships → higher fuel consumption.
- Causes **corrosion** of metals.
- Reduces efficiency of **heat exchangers** and other marine equipment.

Prevention methods:

- **Antifouling coatings** (e.g., paints containing biocides).
- **Regular cleaning** of surfaces.
- Use of **ultrasound or copper surfaces** to discourage attachment.

UNIT-I

2. Describe the interrelations of microbial populations and community dynamics.?

Ans- Microbial populations rarely exist in isolation; instead, they form **complex communities** in various ecosystems, including soil, water, human gut, and extreme environments. The structure, function, and dynamics of these communities are governed by interactions between microbial populations, environmental factors, and resource availability. Understanding these **interrelations** is crucial for ecology, biotechnology, and health.

1. Types of Microbial Interactions

Microbial interactions can be classified based on their ecological relationships:

1.1 Mutualism (+/+)

- Both species benefit.
- Example:
 - *Rhizobium* spp. fix nitrogen for legumes, receiving carbohydrates in return.
 - Gut microbes produce vitamins (e.g., vitamin K) for humans while feeding on undigested food.

1.2 Commensalism (+/0)

- One species benefits, the other is unaffected.
- Example: *Staphylococcus epidermidis* on human skin uses nutrients but does not harm the host.

1.3 Competition (-/-)

- Microbes compete for limited resources such as nutrients, space, or oxygen.
- Competitive exclusion principle: one species may dominate, suppressing others.
- Example: Soil bacteria competing for carbon sources.

1.4 Amensalism (-/0)

- One species is inhibited or destroyed; the other is unaffected.
- Example: *Penicillium* produces penicillin inhibiting bacteria.

1.5 Predation/Parasitism (+/-)

- One organism benefits at the expense of the other.
- Example: Bacteriophages infect and lyse bacteria; protozoa feed on bacteria.

2. Community Dynamics

Microbial communities are dynamic and change in response to environmental conditions, nutrient availability, and interactions within populations.

2.1 Succession

- Sequential changes in community composition over time.
- **Primary succession:** colonization of sterile habitats (e.g., lava rock colonized by cyanobacteria).
- **Secondary succession:** changes after disturbance (e.g., antibiotics altering gut microbiota).

2.2 Stability and Resilience

- Stability: ability to maintain structure under stress.
- Resilience: ability to recover after disturbance.
- Microbial interactions such as mutualism and functional redundancy contribute to resilience.

2.3 Keystone Species

- Certain microbes have disproportionate effects on community structure.
- Example: *Faecalibacterium prausnitzii* in the gut helps maintain microbial balance and anti-inflammatory environment.

3. Mechanisms of Microbial Interrelations

3.1 Metabolic Cooperation

- Cross-feeding: one species uses metabolites produced by another.
- Example: In wastewater biofilms, fermentative bacteria produce short-chain fatty acids consumed by methanogens.

3.2 Quorum Sensing

- Microbes communicate via chemical signals to coordinate behavior, like biofilm formation or virulence.
- Example: *Pseudomonas aeruginosa* uses acyl-homoserine lactones for biofilm maturation.

3.3 Horizontal Gene Transfer

- Genes, including antibiotic resistance or metabolic genes, can spread among populations, shaping community function.
- Mechanisms: transformation, transduction, conjugation.

3.4 Biofilm Formation

- Microbial communities often form biofilms—structured assemblies embedded in extracellular polymeric substances.
- Advantages: protection from stress, enhanced nutrient access, and close interspecies interactions.

4. Environmental Factors Influencing Community Dynamics

1. **Nutrient Availability:** Determines which populations dominate.
2. **pH, temperature, oxygen levels:** Select for specific metabolic types (aerobes, anaerobes, extremophiles).
3. **Disturbance:** Antibiotics, pollution, or grazing can shift population balance.
4. **Spatial Structure:** Microbial niches (surface vs. planktonic) affect interactions.

5. Examples of Microbial Community Dynamics

- **Human Gut Microbiota:** Firmicutes and Bacteroidetes maintain balance through competition and cross-feeding; diet or antibiotics can disrupt this, leading to dysbiosis.

- **Soil Microbiomes:** Nitrogen-fixing bacteria, decomposers, and mycorrhizal fungi interact to recycle nutrients.
- **Marine Microbial Loops:** Cyanobacteria, heterotrophic bacteria, and viruses form complex networks influencing carbon cycling.

6. Implications of Microbial Interrelations

- **Ecological:** Maintain nutrient cycles and ecosystem stability.
- **Medical:** Dysbiosis can cause diseases like IBD, obesity, or infections.
- **Industrial/Biotechnological:** Mixed cultures are used in fermentation, wastewater treatment, and bioremediation.

Conclusion

Microbial populations are interdependent, forming dynamic communities shaped by cooperation, competition, communication, and environmental pressures. Studying these interrelations provides insights into ecosystem function, human health, and biotechnological applications. The balance between interactions—mutualistic, competitive, predatory—is crucial for community stability and resilience.

UNIT-II

3. Write a short note on the microbial production of

(a) Methyl mercury

(b) Tri-methyl arisen

(c) H₂S

Ans- Microbial Production of Toxic and Gaseous Compounds

Microorganisms play a significant role in transforming elements into toxic or volatile forms through metabolic processes. These transformations can have environmental and health impacts.

(a) Methyl Mercury (CH₃Hg⁺)

- **Definition:** Methyl mercury is an organic, highly toxic form of mercury that bioaccumulates in aquatic food chains.

- **Microbial Role:** Certain anaerobic bacteria, particularly **sulfate-reducing bacteria** (e.g., *Desulfovibrio* spp.), can methylate inorganic mercury (Hg^{2+}) into methyl mercury.
- **Mechanism:**
 1. Mercury binds to bacterial enzymes or thiol groups.
 2. Methylation occurs via transfer of a methyl group from organic substrates to Hg^{2+} .
 3. Resulting CH_3Hg^+ is lipophilic and accumulates in organisms.
- **Significance:** Methyl mercury is neurotoxic, causing biomagnification in fish and humans.

(b) Tri-methyl Arsenic ($\text{As}(\text{CH}_3)_3$)

- **Definition:** Tri-methyl arsenic is a volatile, organoarsenic compound produced by microbial methylation of inorganic arsenic.
- **Microbial Role:** Arsenic-methylating bacteria, such as **Methanosarcina** spp. and some **Clostridium** species, can convert arsenate (As^{5+}) or arsenite (As^{3+}) into methylated forms.
- **Mechanism:**
 1. Arsenic detoxification pathway in microbes involves sequential methylation steps.
 2. Inorganic arsenic $\rightarrow$ mono-, di-, and tri-methylated arsenic.
 3. Tri-methyl arsenic is volatile and released into the environment.
- **Significance:** Volatile arsenic compounds contribute to environmental arsenic cycling and pose toxicity risks.

(c) Hydrogen Sulfide (H_2S)

- **Definition:** H_2S is a toxic, colorless gas with a characteristic rotten-egg odor, produced biologically via sulfur metabolism.
- **Microbial Role:** Anaerobic bacteria, such as **Desulfovibrio**, **Clostridium**, and some **Enterobacteriaceae**, produce H_2S by

reducing sulfate or metabolizing sulfur-containing amino acids (cysteine, methionine).

- **Mechanism:**

1. **Sulfate Reduction:** $\text{SO}_4^{2-} \rightarrow \text{SO}_3^{2-} \rightarrow \text{S}^{2-} \rightarrow \text{H}_2\text{S}$
2. **Amino Acid Degradation:** Cysteine $\rightarrow$ pyruvate + NH_3 + H_2S

- **Significance:**

- H_2S contributes to corrosion, malodor, and toxicity in wastewater systems.
- Plays a role in microbial sulfur cycling in nature.

References/Notes:

- Methylation of heavy metals and metalloids is primarily a microbial detoxification strategy.
- Anaerobic conditions favor these transformations.
- Bioaccumulation and volatility are key environmental concerns.

UNIT-IV

5. Write about the role of microorganisms in the degradation of oil spills and pesticides.?

Ans-Role of Microorganisms in Degradation of Oil Spills and Pesticides

Microorganisms, including bacteria, fungi, and some algae, play a crucial role in the **bioremediation** of environmental pollutants such as oil spills and pesticides. They help in breaking down complex organic compounds into simpler, non-toxic substances, thereby reducing environmental damage.

1. Microbial Degradation of Oil Spills

Oil spills are mixtures of hydrocarbons (alkanes, aromatics, resins, and asphaltenes) that are toxic and persistent in the environment. Certain microorganisms can utilize these hydrocarbons as **sources of carbon and energy**, transforming them into harmless products like water and carbon dioxide.

Mechanism:

1. **Enzymatic Breakdown:** Microbes secrete enzymes such as oxygenases and dehydrogenases, which oxidize hydrocarbons.
2. **Biofilm Formation:** Some bacteria form biofilms around oil droplets, enhancing the degradation process.
3. **Aerobic vs Anaerobic Degradation:**
 - Aerobic microbes use oxygen to break down hydrocarbons efficiently.
 - Anaerobic microbes degrade hydrocarbons in oxygen-deficient environments through sulfate or nitrate reduction.

Examples of Oil-Degrading Microorganisms:

- *Pseudomonas*, *Alcanivorax*, *Bacillus* species (bacteria)
- *Candida* and *Aspergillus* species (fungi)

Applications:

- Natural attenuation (using native microbes)
- Biostimulation (adding nutrients to enhance microbial activity)
- Bioaugmentation (introducing oil-degrading microbes)

2. Microbial Degradation of Pesticides

Pesticides, including organophosphates, organochlorines, and carbamates, are toxic chemicals used in agriculture. They can persist in soil and water, harming ecosystems. Microorganisms can detoxify these compounds through enzymatic reactions.

Mechanism:

1. **Hydrolysis:** Microbial enzymes like esterases or hydrolases break pesticide bonds.
2. **Oxidation and Reduction:** Microbes use oxidoreductases to degrade complex pesticide molecules.
3. **Mineralization:** Complete conversion of pesticides to CO₂, H₂O, and inorganic salts.

Examples of Pesticide-Degrading Microorganisms:

- *Bacillus*, *Pseudomonas*, *Enterobacter* species for organophosphates
- *Trichoderma* and *Aspergillus* species for organochlorines

Applications:

- Treating contaminated soils and water
- Reducing pesticide residues in crops through microbial formulations

3. Advantages of Microbial Bioremediation

- Eco-friendly and sustainable
- Cost-effective compared to chemical methods
- Can target a wide range of pollutants
- Can completely mineralize toxic compounds

4. Limitations

- Dependent on environmental factors like temperature, pH, and oxygen
- Some pollutants are recalcitrant and degrade slowly
- Risk of secondary pollution if genetically modified microbes are used

Conclusion

Microorganisms are indispensable agents in the natural detoxification of oil spills and pesticide-contaminated environments. By metabolizing hydrocarbons and pesticides, they reduce pollution, protect ecosystems, and offer an environmentally friendly solution for waste management. Modern biotechnological approaches, including biostimulation and bioaugmentation, have further enhanced their efficiency, making microbial degradation a cornerstone of sustainable environmental remediation.